AFF–1149

Sub. Code
4MCI1C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science and Information Technology

PRINCIPLES OF INFORMATION TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Write the components of a digital computer.

2. Define Interactivity.

3. What is a spreadsheet?

4. Write the types of application software.

5. What is the difference between analog and digital signal?

6. Define Modem.

7. What is file management?

8. State any two features of DBMS.

9. Write the various generations of programming languages.

10. What is Information super highway?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the elements of a communication system.

Or

(b) Discuss about the ethics of information technology.

12. (a) Explain the features of word processor.

Or

(b) Write the difference between database software and presentation graphics software.

13. (a) Discuss in detail about telephone related communication services with diagram.

Or

(b) How communications are done through channels? Explain with a neat sketch.

14. (a) Write and explain the various types of secondary storage devices.

Or

(b) Explain briefly about the types of database organization.

15. (a) Discuss about management information systems.

Or

(b) Explain the tools available in visual programming.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about the development of computer and communication technology.

17. Briefly explain about the tools available for thinking and working in application software.

18. What are ISDN lines? Explain its uses to make communication with a neat diagram.

19. Explain about data organization in secondary storage devices.

20. Explain the phases of system analysis design in detail.

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AFF–1150

Sub. Code
4MCI1C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science and Information Technology

C AND DATA STRUCTURE

(CBCS – 2014 onwards)

Time : Three Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a static variable?

2. State the purpose of main ( ) function.

3. What is use of if statement?

4. Write the syntax of while loop.

5. What is an array?

6. Mention any two advantages of functions.

7. List out the areas in which data structures are applied extensively.

8. Differentiate between a queue and a stack.

9. What are complete trees?

10. What do you meant by insertion sort?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Enlist the features of C.

Or

(b) Explain any two bitwise operators with suitable example.

12. (a) Write a note on Nested switch ( ) case statement with example.

Or

(b) Explain about Nested for loop with example.

13. (a) Narrate about the features of Pointers.

Or

(b) Discuss about Two dimensional array with suitable example.

14. (a) What is Data Structure? Briefly explain various types of Data Structure.

Or

(b) Write an algorithm to insert and delete a node in Singly Linked List.

15. (a) Explain about Binary search with example.

Or

(b) Discuss about Bubble sort with suitable example.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain different data types used in C language.

17. Discuss about continue and break statements with suitable example.

18. Explain any five string functions with suitable example.

19. What is Stack? Write the algorithm of PUSH and POP operation on Stack.

20. Discuss about Inorder, Preorder and Postorder Traversal operations on Binary tree with appropriate example.

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Sub. Code
4MCI1C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science and Information Technology

JAVA PROGRAMMING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by 'static' in Java?

2. List any two features of Java.

3. How to define an Interface?

4. Why multiple inheritance using classes is disadvantage in Java?

5. What method must be implemented by all threads?

6. Why do threads block on I/O?

7. What is an applet? How do applets differ from application programs?

8. How get Parameter( ) method is used in applet?

9. What is a package class?

10. Define socket.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Discuss the various types of operators available in Java. Give examples.

Or

(b) Write a Java program to find the sum of given numbers using command line arguments.

12. (a) Explain about abstract class.

Or

(b) Write short notes on User defined exceptions.

13. (a) Discuss about multi threading.

Or

(b) Make a note on Interthread communication.

14. (a) Write short notes on string handling in Java.

Or

(b) Explain the steps involved to execute and run an Applet.

15. (a) Discuss about URL.

Or

(b) Write short notes on JSP.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain about Java Development Environment in detail.

17. Define an exception and explain how the exceptions are handled.

18. Explain the different states of a thread.

19. What is Event Handling in Java? List the available event classes and listener interfaces with suitable example.

20. Write a detailed note on JDBC.

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AFF–1152

Sub. Code
4MCI1C4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science and Information Technology

DIGITAL COMPUTER FUNDAMENTALS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

All questions carry equal marks.

1. What is J-K flip-flop?
2. What is X-NOR gate?
3. What is the purpose of ALU?
4. Write down 8 to 1 line multiplexer.
5. Define PROM.
6. Differentiate RAM and ROM.
7. Define Bus .
8. What is memory mapped I/O?
9. Specify the role of control word.
10. Mention the function of shift instructions.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

All questions carry equal marks.

11. (a) Draw logic diagram for following expressions.

    (i) AB+CD

    (ii) Y=AB+ (B+C).

Or

(b) What is flip flop? Explain-S flip flop with truth table?

12. (a) Explain with block diagram 2 - 1 line multiplexer.

Or

(b) Write short notes on binary adder.

13. (a) Describe in detail about the Arithmetic operations using Adder and logical operations.

Or

(b) Discuss briefly about the importance of virtual memory.

14. (a) Discuss in detail about the codes for detection and correction of errors.

Or

(b) Write short notes on Serial transmission of character codes.

15. (a) Write short notes on control register.

Or

(b) Write short notes on microprogramming.

Part C (3 × 10 = 30)

Answer any **three** questions.

All questions carry equal marks.

16. Describe the integrated design and various integrated circuits in details.

17. Explain in detail about partially binary adder with neat sketch.

18. Discuss about the Virtual and Cache memories.

19. Explain In details about Interfacing of a Keyboard and a Printer.

20. How to construct an instruction and control registers?

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AFF–1154

Sub. Code
4MCI1E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

First Semester

Computer Science and Information Technology

***Elective* — COMPUTER SYSTEM ARCHITECTURE**

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (10 × 2 = 20)

Answer **all** questions.

1. Define a stack.

2. What do you mean by Direct Addressing?

3. What is the use of Arithmetic operation?

4. State any two Control functions.

5. What is meant by an Microprogram?

6. Write any two functions of Microprogram Control.

7. What is meant by Priority Interrupts?

8. State any two peripheral devices.

9. What is Pipelining?

10. Define Asynchronous Data transfer.

Section B (5 × 5 = 25)

Answer **all** questions.

11. (a) Discuss brief notes on ALU with its functions.

Or

(b) Describe briefly about Program Control.

12. (a) Explain briefly the Basic Computer Organization.

Or

(b) Describe briefly about the Output Interrupt.

13. (a) Write Short notes on Microprogram Addressing and Sequencing.

Or

(b) Describe briefly the advantages of Microprogram.

14. (a) Explain briefly the Synchronous Data transfer.

Or

(b) Describe briefly about Virtual Memory.

15. (a) Explain briefly the Arithmetic Pipelining Process.

Or

(b) Describe briefly about Array processing.

Section C (3 × 10 = 30)

Answer any **three** questions.

16. Write brief notes on different addressing modes.

17. Explain the various logical shift micro operations.

18. Discuss in detail about Micro Instruction Formats.

19. Explain about operations of Cache Memory.

20. Discuss in detail about the Instruction pipelining.

AFF–1155

Sub. Code
4MCI2C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science and Information Technology

DATABASE TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is a data model? List the types of data models used.

2. What are the pitfalls in relational database design?

3. What are aggregate functions? And list the aggregate functions supported by SQL?

4. How might the type of index available influence the choice of a query processing strategy?

5. What are ACID properties?

6. When is a transaction said to be deadlocked?

7. State the advantages of Distributed Database.

8. Give at least two differences between Mobile and Web databases.

9. What are the benefits of data warehouse?

10. What is meant by security threats?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Describe the components of DBMS.

Or

(b) Discuss on various operations involved in Relational Algebra.

12. (a) Write a note on SQL facilities.

Or

(b) Explain the steps involved in query processing.

13. (a) Describe two phase commit protocol with examples.

Or

(b) With a neat diagram explain the states of a transaction.

14. (a) Distinguish between Multidimensional and Parallel databases.

Or

(b) Discuss on Spatial and multimedia databases.

15. (a) Explain the architecture of Data warehouse in detail with a neat diagram.

Or

(b) Define Data Mining. Explain the steps followed in Mining process.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the normalization techniques using Multi Valued Dependency with example.

17. Explain Query processing with examples to perform sort and join operation.

18. Explain different locking mechanisms used in lock based concurrency control.

19. Discuss on Distributed Database.

20. Describe the Statistical Database Security.

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AFF–1156

Sub. Code
4MCI2C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science and Information Technology

VISUAL PROGRAMMING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define data type.

2. State the need of Control Array.

3. What is an Active X Control?

4. Write the difference between list box and combo box control.

5. State the features of MDI.

6. What is Rich Edit control?

7. Define Data report.

8. What is the use of ADODC?

9. Write the uses of message maps.

10. What is the application of MDI?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Distinguish between procedures and functions in Visual Basic and develop a Visual Basic Program for a banking system.

Or

(b) Explain various control flow statements in VB with example.

12. (a) Explain the steps involved in adding a tool bar control to a Visual Basic interface.

Or

(b) State the advantages of Active X Control and Write a program to create a single Active X control.

13. (a) Discuss on the Rich Text Box control with an example.

Or

(b) How files are supported by Visual Basic? Explain it.

14. (a) Compare ADO and DAO.

Or

(b) Discuss Data Environment in ADODC.

15. (a) Explain the various common dialogs available in VC++.

Or

(b) What is Document View Architecture? Compare it with an ordinary application.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the various loop statements in Visual Basic and Write a simple Visual Basic program for each statement.

17. (a) Write a Visual Basic program that handles the click event of the right move button with a "Welcome" message and explain it. (5)

    (b) Explain indexing List Box Control. (5)

18. Discuss about Advanced Active X Control with its properties and methods.

19. Design a Library Management System with database connectivity. Give the steps involved in achieving the data base connectivity with anyone of the controls provided by Visual Basic.

20. Design a simple calculator to perform arithmetic operations and Write a necessary code to implement it in VC++.

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Sub. Code
4MCI2C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science and Information Technology

COMPUTER NETWORKS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What are the advantages of computer network?

2. Which of the layers in the OSI model are network support layers?

3. How twisted pair cable is different from fiber optic cable?

4. Why switching is needed?

5. What is the difference between even parity and odd parity?

6. Define HDLC.

7. What do you mean by datagram?

8. List out the three main elements of link state routing.

9. What is the rote of ASN.1?

10. Define Virtual Terminal.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Define topology. Explain its types.

Or

(b) What is protocol? What are its key elements? Discuss.

12. (a) List and explain the various types of modulation.

Or

(b) What is meant by polling? Explain its significance.

13. (a) Explain the five key assumptions in dynamic channel allocation.

Or

(b) How single bit error is different from burst error? Discuss with an example.

14. (a) Discuss the important categories of service are useful for describing the transport service.

Or

(b) Write short notes on synchronization points.

15. (a) What are the important objectives of presentation layer? Explain.

Or

(b) List and explain the basic aspects of electronic mail system.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Describe the various services provided by the layers in OSI reference model.

17. Discuss in detail about any three transmission media with neat sketch.

18. Explain briefly about elementary data link protocols with appropriate illustrations.

19. What do you mean by routing? Discuss any two routing algorithms with neat sketch.

20. Write short notes on the following :

    (a) FTAM. (5)

    (b) Virtual Terminals. (5)

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Sub. Code
4MCI2E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science and Information Technology

Elective — COMPUTER ORIENTED NUMERICAL METHODS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. State the condition of convergence of Newton – Raphson method.

2. What is the basic principle using which Horner's method has been established?

3. What is the difference between triangular factorization and Crout's reduction methods?

4. How does the coefficient matrix A in the system $AX = B$ get transformed in Gauss elimination method?

5. Define interpolating polynomial.

6. State Gauss's forward interpolation formula.

7. Write down the formula for $\frac{dy}{dx}$ at any point in terms of Newton's backward interpolation formula.

8. What do you mean by numerical integration?

9. State Picard's iteration formula.

10. Find $y(1.1)$ using Euler's method, Given $\frac{dy}{dx} = 2x - y$ and $y(1) = 3$.

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Use Regula Falsi method to find the real root of the equation $x^3 - x^2 - 2 = 0$.

Or

(b) Using Newton – Raphson method find the root of the equation $3x - \cos x - 1 = 0$.

12. (a) Solve by Gauss elimination method $3x - y + 2z = 12; x + 2y + 3z = 11$; $2x - 2y - z = 2$.

Or

(b) Solve by Gauss – Seidel iterative method $27x + 6y - z = 85$; $6x + 15y + 2z = 72$;

$x + y + 54z = 110$.

13. (a) Find y when $x = 27$ from the following data

x :	10	15	20	25	30
y :	35.4	32.2	29.1	26.0	23.1

Or

(b) Use Lagrange's formula to find the value of y at $x = 6$ from the following data

x :	3	7	9	10
y :	168	120	72	63

14. (a) From the following data find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$ at $x = 1.05$

x :	1.00	1.05	1.10	1.15	1.20	1.25
y :	1.0000	1.0247	1.0488	1.0724	1.0954	1.1180

Or

(b) Evaluate $\int_0^1 \frac{1}{1+x} dx$, dividing the range into eight equal parts using Simpson's rule.

15. (a) Use Taylor series method to find $y(0.1)$ given that $\frac{dy}{dx} = x^2 - y; y(0) = 1$.

Or

(b) Use Runge – Kutta method find $y(0.1)$ given that $\frac{dy}{dx} = x + y; y(0) = 1$

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Find by Horner's method the root of the equation $x^3 + x^2 + x - 100 = 0$.

17. Solve the following system by the method of triangularisation.

$2x - 3y + 10z = 3; x - 4y - 2z = -20; 5x + 2y + z = -12$.

18. Apply Bessel's formula to find $f(1.95)$ given that

x :	1.7	1.8	1.9	2.0	2.1	2.2	2.3
$f(x)$:	2.979	3.144	3.283	3.391	3.463	3.997	4.491

19. Evaluate $\int_0^1 \frac{dx}{1+x^2}$ by using Romberg's method correct to 4 decimal places.

20. Given $\frac{dy}{dx} = \frac{1}{2}\left(1+x^2\right)y^2$ and $y(0)=1, y(0.1)=1.06$ $y(0.2)=1.12$, $y(0.3)=1.21$. Evaluate $y(0.4)$ by Milne's predictor corrector method.

AFF–1159

Sub. Code
4MCI2E2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Second Semester

Computer Science and Information Technology

Elective – APPLIED MATHEMATICS FOR COMPUTER SCIENCE

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A $(10 \times 2 = 20)$

Answer **all** questions.

1. Construct truth table for $P \leftrightarrows (Q \vee P)$.

2. Show that $\neg(P \to Q) \Rightarrow P$.

3. Express $P \uparrow Q$ interms of $\downarrow$ only.

4. Define Maxterm and Minterm.

5. How many edges are there in a graph with 10 vertices each of degree 5?

6. Define regular graph.

7. Define redundant constraint.

8. How the LPP has unbounded solution is identified in the simplex table?

9. What do you mean by transportation problem.

10. Explain how maximization problems are solved using assignment model technique?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Prove that $P \to (Q \vee R) \Leftrightarrow (P \to Q) \vee (P \to R)$.

Or

(b) Show that $(P \to Q) \wedge (Q \to R) \Rightarrow P \to R$

12. (a) Prove that $\{\uparrow\}$ is functionally complete set.

Or

(b) Show that

(i) $\rceil(P \uparrow Q) \Leftrightarrow \rceil P \downarrow \rceil Q$

(ii) $\rceil(P \downarrow Q) \Leftrightarrow \rceil P \uparrow \rceil Q$

13. (a) Draw the graph with the following adjacency matrix.

$$\begin{pmatrix} 1 & 2 & 0 & 1 \\ 2 & 0 & 3 & 0 \\ 0 & 3 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{pmatrix}$$

Or

(b) Represent the expression $((a+b)-((c*d)-(e/f)))*a$ as a binary tree and write the prefix and postfix forms of the expression.

14. (a) Solve by graphical method;
max $z = 5x_1 + 8x_2$

subject to $15x_1 + 10x_2 \le 180$
$10x_1 + 20x_2 \le 200$
$15x_1 + 20x_2 \le 210$
and $x_1, x_2 \ge 0$

Or

(b) Solve by simplex method to:
max $z = 3x_1 + 4x_2$

subject to $2x_1 + 5x_2 \le 120$
$4x_1 + 2x_2 \le 80$
and $x_1, x_2 \ge 0$

15. (a) Find the initial basic feasible solution for the following transportation problem using least cost method.

	F_1	F_2	F_3	F_4	Availability
D_1	11	13	17	14	250
D_2	16	18	14	10	300
D_3	21	24	13	10	400
Requirement	200	225	275	250	

Or

(b) Solve the following assignment problem

	J_1	J_2	J_3	J_4	J_5
M_1	9	22	58	11	19
M_2	43	78	72	50	63
M_3	41	28	91	37	45
M_4	74	42	27	49	39
M_5	36	11	57	22	25

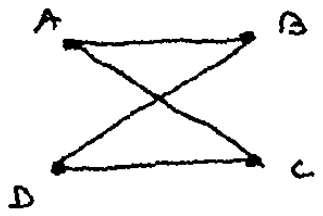
Part C (3 × 10 = 30)

Answer any **three** questions.

16. Obtain the PCNF and PDNF of $(P \to (Q \wedge R)) \wedge (\neg P \to (\neg Q \wedge \neg R))$.

17. Show that SVR is tautologically implied by $(P \vee Q) \wedge (P \to R) \wedge (Q \to S)$.

18. Find the number of paths of length 4 from the vertex A to D in the simple graph analytically



19. Use simplex method to

Maximize $Z = 2x_1 + x_2 + x_3$

Subject to $4x_1 + 6x_2 + 3x_3 \le 8$

$3x_1 - 6x_2 - 4x_3 \le 1$

$2x_1 + 3x_2 - 5x_3 \ge 4$

and $x_1, x_2, x_3 \ge 0$

20. Solve the following transportation problem

						Supply
	4	1	2	6	9	100
	6	4	3	5	7	120
	5	2	6	4	8	120
Demand	40	50	70	90	90	

AFF–1163

Sub. Code
4MCI3C1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science and Information Technology

PRINCIPLES OF COMPILER DESIGN

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define compiler.

2. What are the three general approaches to the implementation of a Lexical analyzer?

3. What is meant by symbol table?

4. What is Hash table?

5. What are the methods of representing a syntax tree?

6. Give the syntax directed definition of if-else statement.

7. What are the limitations of storage allocation?

8. What are the four kinds of intermediate code often used in compilers?

9. Write down the issues of code generator.

10. Define DAG.

Part B $(5 \times 5 = 25)$

Answer **all** questions choosing either (a) or (b).

11. (a) Discuss the importance of input buffering in the Lexical analyzer.

Or

(b) How identifiers and keywords can be recognized using Lexical analyzer.

12. (a) Discuss on top down parsing in briefly.

Or

(b) What is meant by left recursion? Explain and eliminate it with an example.

13. (a) Write short note on adapting translation schemes.

Or

(b) Discuss about the specification of a simple type checker.

14. (a) Briefly explain about parameter passing in storage organization.

Or

(b) What are the methods of translating Boolean expressions?

15. (a) Discuss about the importance of code optimization.

Or

(b) Discuss about the importance of flow graphs.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail about the different phases of compiler with neat diagram.

17. Give an algorithm for constructing an SLR parsing table, Explain the steps with an example.

18. What do you mean by syntax directed translation scheme? Discuss some of its popular forms and their implementation.

19. Describe about the various source language issues in briefly.

20. Discuss run time storage management issues in briefly.

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AFF–1164

Sub. Code
4MCI3C2

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science and Information Technology

SOFTWARE ENGINEERING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** the questions.

1. Define Process.

2. What are the limitations of Spiral Model?

3. What is the purpose of DFD?

4. Define SRS.

5. Why Estimation is so important?

6. List out the major activities of Risk Management.

7. What is the difference between Functional and Data abstraction?

8. Mention the various levels of Cohesion.

9. Define Failure.

10. What are the two basic approaches of Testing?

Part B $(5 \times 5 = 25)$

Answer **all** questions, choosing either (a) or (b).

11. (a) What are the different types of Software Quality Attributes? Discuss.

Or

(b) Explain the key outputs in a development project that follows the Prototype Model.

12. (a) Write short notes on Data Dictionary.

Or

(b) Define Prototyping. Explain its importance.

13. (a) Discuss the essential guidelines for Software Project Planning.

Or

(b) What are the limitations of Cost Estimation Models? Explain.

14. (a) Explain the importance of Design Process.

Or

(b) What is the difference between a Flow Chart and a Structure Chart? Discuss.

15. (a) Mention and explain the essential attributes of a Good Test.

Or

(b) What errors are commonly found during Unit Testing? Discuss.

Part C

(3 × 10 = 30)

Answer any **three** questions.

16. Describe any three Software Process Models with suitable illustrations.

17. Discuss in detail about Software Metrics with suitable examples.

18. What is meant by project estimation? Explain COCOMO model of estimation in detail.

19. Describe how structural partitioning can help to make software more maintainable. Justify your answer.

20. Give a detailed note on Black Box Testing with suitable illustrations.

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AFF–1165

Sub. Code
4MCI3C3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science and Information Technology

.NET TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is the basic functionality of Garbage Collector in .Net?

2. Define Common Language Specification (CLS).

3. What is TRACE in VB.Net?

4. Write the difference between Check box and Combo box.

5. How do you exclude an ASP.NET page from using Themes?

6. Differentiate between ASP and ASP.NET.

7. What is overriding?

8. Define abstraction.

9. Which namespaces are required to enable the use of databases in ASP.NET pages?

10. List any two benefits of using of ADO.NET.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write short note on CLR.

Or

(b) What are the main components of .NET Framework? Explain.

12. (a) Explain how exceptions are handled in VB.Net.

Or

(b) Make a note on Toolbar and Status bar.

13. (a) Summarize about file types of ASP.Net.

Or

(b) Discuss about state management in ASP.Net.

14. (a) Explain the concept of shadowing.

Or

(b) Narrate about Form authentication.

15. (a) Write a note on data grid in ADO.Net.

Or

(b) Give a brief account on data objects in ADO.Net.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Describe .NET framework with suitable illustration.

17. Explain about subprocedures and functions in VB.Net.

18. Discuss in detail about HTML server controls in ASP.Net.

19. Give a detailed account on the concept of encapsulation with suitable example.

20. Write a detailed note on ADO.Net.

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AFF–1166

Sub. Code
4MCI3E1

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science and Information Technology

Elective – MULTIMEDIA AND ITS APPLICATIONS

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Define the term Multimedia.

2. What is multimedia software?

3. Define multimedia system.

4. State any two audio file formats.

5. Define sound.

6. What are the components of JPEG?

7. What is multimedia conferencing?

8. Write down the basic requirements for multimedia communications.

9. List down most common protocols used on the Internet.

10. State the function of project manager.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Briefly explain the history and future of multimedia.

Or

(b) Discuss on multimedia computer components.

12. (a) Explain the different file formats used in multimedia.

Or

(b) Describe the elements of text and graphics.

13. (a) Explain how Images can be obtained for multimedia.

Or

(b) List and explain important steps and considerations in recording and editing digital audio.

14. (a) Explain in detail teleconferencing systems.

Or

(b) Discuss on network services and protocols.

15. (a) Briefly discuss the requirements of multimedia project.

Or

(b) Describe anyone sample projects of multimedia application development.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Discuss in detail about the multimedia interface standards.

17. Explain Multimedia system services architecture.

18. Explain the different techniques used for video compression.

19. Describe the multimedia conferencing architecture.

20. Explain the process involved in multimedia project planning and costing.

AFF–1168

Sub. Code
4MCI3E3

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science and Information Technology

Elective — WEB TECHNOLOGY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is CSS?

2. State the use of <FONT> tag.

3. How and where XML can be used?

4. What do you meant by Dynamic HTML?

5. How do you convert numbers between different bases in JavaScript?

6. What is a way to append a value to an array in JavaScript?

7. In how many ways you can retrieve the data in the result set of MySQL using PHP?

8. What is Implode( ) function in PHP?

9. Which feature of Perl provides code reusability?

10. Write the syntax to add two arrays together in Perl.

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Explain the three main ways to apply CSS styles to a Web page.

Or

(b) How do you use color and images in HTML? Explain.

12. (a) Write short notes on Essential AJAX.

Or

(b) Make a note on basics of XML.

13. (a) Summarize about JavaScript basics.

Or

(b) Explain how String is manipulated in JavaScript.

14. (a) Discuss about Server-side Web Scripting.

Or

(b) How do you display queries in tables? Explain.

15. (a) Explain about Scalars in Perl.

Or

(b) Write short notes on regular expressions in Perl.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the features of HTML in detail.

17. Briefly discuss the principles of AJAX.

18. Give a detailed note on arrays in JavaScript with suitable example.

19. Discuss in detail about MySQL Database administration.

20. Explain how to handle XML with Perl in detail.

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AFF–1169

Sub. Code
4MCI3E4

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science and Information Technology

Elective : DATA MINING AND WAREHOUSING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. Compare Data Mining Vs query tools.

2. List out the data mining tasks.

3. What is Data cube?

4. Define OLAP.

5. What is data reduction?

6. What is the need of multidimensional data?

7. How prediction is different from classification?

8. What is outlier analysis?

9. Define Graph mining.

10. What is cluster analysis?

Part B (5 × 5 = 25)

Answer **all** questions.

11. (a) Discuss about data mining techniques.

Or

(b) Briefly explain the term "Machine Learning".

12. (a) Describe in detail about OLAP operations.

Or

(b) Explain development life cycle.

13. (a) Describe the steps involved in Data pre-processing.

Or

(b) Explain the features of Query language.

14. (a) Discuss about Grid Based Method.

Or

(b) Explain about Bayesian classification.

15. (a) Discuss about partitioning method.

Or

(b) Explain the features of Time series and sequence mining.

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain the steps in data mining process.

17. Explain the features of Data Warehouse and Data Warehouse Architecture.

18. Discuss in detail about Discretization and concept Hierarchy generation.

19. Describe in detail about Back Propagation method.

20. Explain about web mining and web content mining.

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AFF–1170

Sub. Code
4MCI3E5

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science and Information Technology

Elective – INFORMATION SECURITY

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is meant by attack?

2. What is threat?

3. Mention any four threats.

4. Define DDoS attack.

5. State the components of risk identification.

6. Categorize the components of an information system.

7. Mention the components of ISSP.

8. State the components of EISP.

9. How does physical access control differ from the logical access control?

10. Why is it important to use specific and dearly defined job descriptions for hiring information security professionals?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write down the history of information security briefly.

Or

(b) Explain the critical characteristics of information.

12. (a) Discuss the four important functions performed by information security for an organization.

Or

(b) Describe the law and ethics in information security.

13. (a) Give an overview of risk management.

Or

(b) Briefly discuss the five risk control strategies.

14. (a) Discuss the systems-specific policy.

Or

(b) Explain the policy management.

15. (a) Write down about any five of the major physical security controls.

Or

(b) What are the advantages and disadvantages of HIDPSs?

Part C (3 × 10 = 30)

Answer any **three** questions.

16. Explain in detail the security systems development life cycle.

17. Present the major types of attacks used against controlled systems.

18. Discuss in detail about risk assessment.

19. Describe the design of security architecture.

20. Explain about the scanning and analysis tools.

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AFF–1171

Sub. Code
4MCI3E6

M.Sc. DEGREE EXAMINATION, NOVEMBER 2015

Third Semester

Computer Science and Information Technology

Elective – SOFT COMPUTING

(CBCS – 2014 onwards)

Time : 3 Hours Maximum : 75 Marks

Part A (10 × 2 = 20)

Answer **all** questions.

1. What is neural network?

2. What is activation function?

3. What is feed forward network?

4. Define radial basis function.

5. What is self organizing network?

6. What is Hopfield network?

7. List the basic operations of set.

8. What is equivalence relationship?

9. What is reproduction?

10. What is membership function?

Part B $(5 \times 5 = 25)$

Answer **all** questions.

11. (a) Write the applications of soft computing.

Or

(b) Write short note on Mc Colloch Pitts Neuron.

12. (a) Write short notes on Madaline.

Or

(b) Discuss in detail about Radial Basis function network.

13. (a) Write short notes on Hebbian learning.

Or

(b) Explain BAM with a neat diagram.

14. (a) Explain why we need Fuzzy set theory.

Or

(b) Discuss the feature of membership function.

15. (a) Explain mutation operation with example.

Or

(b) Explain genetic programming.

Part C $(3 \times 10 = 30)$

Answer any **three** questions.

16. Explain in detail the fundamental concepts of ANN.

17. Explain in detail about BPN with neat diagram.

18. Discuss in detail about Adaptive Response Theory.

19. Explain in detail the concepts of Fuzzy relations.

20. Explain hybrid genetic algorithm and its application.

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